

Unit - 1 SAG:-

1) Sum of first "n" natural numbers is $S_n = \frac{n(n+1)}{2}$

$$S_{20} = \frac{20(21)}{2} = \frac{420}{2} = 210$$

$\therefore$ Sum of first 20 natural numbers is 210

2) Sum of $1^2 + 2^2 + 3^2 + \dots + 10^2$

Sum of first natural square numbers is

$$S_n = \frac{n(n+1)(2n+1)}{6} \quad \text{where } n = 10$$

$$= \frac{10(10+1)(2(10)+1)}{6} = \frac{10(11)(21)}{6}$$

$$= \frac{2310}{6} = \underline{\underline{385}}$$

3) $167 \times 2183 \times 497 \times 839 \times 235 \times 111 \times 1039 \times 251 \times 563$

To get the value of unit digit we only need to

Consider unit digits of individual number

$$\Rightarrow \cancel{7 \times 3} \times \cancel{7 \times 9} \times \cancel{5 \times 1} \times \cancel{9 \times 1} \times \cancel{3}$$

$$\Rightarrow 21$$

$$\Rightarrow 7 \times 3 = 21$$

$$1 \times 7 = 7$$

$$7 \times 9 = 63$$

$$3 \times 5 = 15$$

$$5 \times 1 = 5$$

$$5 \times 9 = 45$$

$$5 \times 1 = 5$$

$$5 \times 3 = 15 \text{ which is "5"}$$

$\therefore$ unit digit will be 5.

4) Remainder of $232 \times 43 \times 2347 \times 4671 \times 6988$

We can calculate remainder by dividing the each individual number and noting the remainder.

$$\Rightarrow \frac{232}{23} \times \frac{43}{23} \times \frac{2347}{23} \times \frac{4671}{23} \times \frac{6988}{23}$$

$$\Rightarrow 2 \times \cancel{20} \times 1 \times \cancel{2} \times 19 \Rightarrow \cancel{2 \times 3 \times 2}$$

$$\Rightarrow \frac{2 \times (-3) \times 1 \times 2 \times (-4)}{23}$$

$$= \frac{6 \times 2 \times 4}{23} \Rightarrow \frac{48}{23} = 2 \text{ (Remainder)}$$

$$\begin{array}{r} 2 \\ 23 \overline{) 43} \\ \underline{-46} \\ -3 \end{array}$$

$$\begin{array}{r} 304 \\ 23 \overline{) 6988} \\ \underline{-69} \\ 88 \\ \underline{-87} \\ 1 \end{array}$$

5) $80!$

Number of zeros $\Rightarrow$
$$\begin{array}{r} 5 \overline{) 80} \\ 5 \overline{) 16} \\ 3 \end{array} \} 19.$$

$\therefore 19$ zeros are in $80!$

(or)

$$\Rightarrow \left[\frac{80}{5} \right] + \left[\frac{80}{5^2} \right] + \left[\frac{80}{5^3} \right] + \dots$$

$$\Rightarrow 16 + 3 + 0 = \underline{19}$$

6) $8 \times 88 \times 888$

$$\begin{array}{r} 8 \\ 88 \\ \hline 704 \\ 888 \\ \hline 625152 \end{array}$$

$$\Rightarrow 625,152$$

By prime factorization

7) $16 \Rightarrow 2^4$

$18 \Rightarrow 2 \times 3^2$

$24 \Rightarrow 2^3 \times 3$

$30 \Rightarrow 2 \times 3 \times 5$

$LCM = 2^4 \times 3^2 \times 5$

$\Rightarrow 16 \times 9 \times 5$

$= \underline{720}$

- 8) Given,
The numbers 8, 9, 12, 15 should leave same remainder 5 when divided by 8.

Now,

To find the least number which divides the given numbers and leave 5 as remainder is

$$\Rightarrow \text{LCM}(8, 9, 12, 15)$$

$$8 \Rightarrow 2^3$$

$$9 \Rightarrow 3^2$$

$$12 \Rightarrow 2^2 \times 3$$

$$15 \Rightarrow 3 \times 5$$

$$\text{LCM} = 2^3 \times 3^2 \times 5$$

$$= 8 \times 9 \times 5$$

$$= 360 \text{ (Perfectly divides)}$$

$$\text{Required number is } 360 + 5$$

$$\Rightarrow \underline{\underline{365}}$$

9)
$$\frac{423 \times 423 - 325 \times 325}{423 - 325}$$

This is in the form $\frac{a^2 - b^2}{a - b} \Rightarrow \frac{(a+b) \times (a-b)}{(a-b)}$

Then, $\boxed{\frac{a^2 - b^2}{a - b} = a + b}$

Here,

$$\Rightarrow a = 423 ; b = 325$$

$$\therefore 423 + 325 = \underline{\underline{748}}$$

10)
$$\begin{array}{r} 2 \overline{) 4200} \\ 7 \overline{) 2100} \\ 3 \overline{) 300} \\ 2 \overline{) 100} \\ 2 \overline{) 50} \\ 5 \overline{) 25} \\ 5 \overline{) 5} \\ 1 \end{array}$$

Prime factors are $\Rightarrow 2^3 \times 3^1 \times 5^2 \times 7^1$

Total factors are $\Rightarrow (a+1)(b+1)(c+1)$
where, $a, b, c \dots$ are the powers of prime factors

$$\begin{aligned} \Rightarrow (3+1)(1+1)(2+1)(1+1) &= (4)(2)(3) \\ &= 8 \times 6 \\ &= 48 // \end{aligned}$$

Unit-2 SAQ:-

1) Successive percentage formula is given by

$$x + y + \frac{xy}{100}$$

$$x = +25 \text{ (Increase)}$$

$$y = -12 \text{ (Decrease)}$$

$$\Rightarrow 25 - 12 + \frac{(25)(-12)}{100}$$

$$\Rightarrow 13 + \frac{(-300)}{100}$$

$$= 13 - 3 = \underline{\underline{10\% \text{ Increase}}}$$

2) 4% of 3125 Rs

$$\Rightarrow 3125 \times \frac{4}{100}$$

$$\Rightarrow \frac{12500}{100} = \underline{\underline{125 \text{ Rs}}}$$

3) Given,

Money spent = 69% of total money

Let, Initially money she has = 100%

$$\text{Remaining amount} = 100\% - 69\% = 31\%$$

Now, Money she has now = 93 Rs.

$$31\% \text{ of } x = 93$$

$$\frac{31}{100} \times x = 93 \Rightarrow \frac{93 \times 100}{31} = x$$

$$\cancel{x = 31 \times}$$

$$\Rightarrow \frac{9300}{31} = x$$

$$\boxed{x = 300}$$

4) Given, Increased price of 1kg rice = 25% of initial price

Let, initial price be = P.

and, 25% of P = 12 Rs.

Now, original price = $\frac{25}{100} \times P = 12$

$$P = \frac{1200}{25}$$

$$P = 48 \text{ Rs}$$

The original price is 48

The new price will be $48 + 12 = 60 \text{ Rs.}$

5) Let salary of B be 100%

Now,

A's salary = $100 + 20\%$ of 100

$$\Rightarrow 120$$

Difference in salary is $\Rightarrow 120 - 100 = 20$.

if A is 20% more than B, Then B is less than A

is given as $\left(\frac{20}{100+20} \times 100 \right) \%$

$$\Rightarrow \left(\frac{20}{100+20} \times 100 \right) \%$$

$$\Rightarrow \left(\frac{20}{120} \times 100 \right) \%$$

$$\Rightarrow \frac{200}{12} = 16.67\%$$

$\therefore$ B's salary is 16.67% less than A.

$$6) \text{ Profit percentage} = \left(\frac{\text{Profit}}{\text{Cost Price}} \right) \times 100$$

$$= \left(\frac{4}{20} \right) \times 100$$

$$= \frac{1}{5} \times 100 = 20\%$$

$$7) \text{ Cost of Typewriter} = 1200$$

$$\text{Extra Cost to repairs} = 200$$

$$\text{Total cost of Typewriter} = 1400$$

$$\text{Profit} = \text{SP} - \text{CP}$$

$$= 1680 - 1400$$

$$= 280 \text{ Rs.}$$

$$\text{Profit \%} = \left(\frac{\text{Profit}}{\text{CP}} \right) \times 100$$

$$= \left(\frac{280}{1400} \right) \times 100$$

$$\Rightarrow \frac{280}{14} = 20\% \text{ profit}$$

8) ~~let~~

$$\text{Gain} = \text{SP} - \text{CP}$$

$$= 1000 - 900$$

$$\text{Gain} = 100 \text{ units}$$

$$\text{Gain \%} = \left(\frac{\text{True weight} - \text{False weight}}{\text{False weight}} \right) \times 100$$

$$= \left(\frac{1000 - 900}{900} \right) \times 100$$

$$= \left(\frac{100}{900} \right) \times 100 = \frac{100}{9} = 11.11\%$$

9) 2%, 4%, 6%, 8%, ...

50 successive discounts

This is in an successive A.P. $\Rightarrow 2, 4, 6, 8, \dots$

The n^{th} discount is $\Rightarrow 50 \times 2 = 100\%$

10) Given,

Discount 1 $\Rightarrow 10\%$

Discount 2 $\Rightarrow 15\%$

Successive discounts $\Rightarrow 10 + 15 - \frac{10(15)}{100}$

$$25 - \frac{150}{100}$$

$$\Rightarrow 25 - 1.5 = 23.5\%$$

Expt-3 SAG:-

1) Given,

$P = 800 \text{ Rs}$

$R = 18\%$

$T = 1 \text{ year}$

$$S.I = \frac{P \times T \times R}{100}$$

$$\Rightarrow \frac{800 \times 1 \times 18}{100}$$

$$\Rightarrow 18 \times 8$$

$$S.I = 144 \text{ Rs}$$

2) Given,

$P = 4000$

$T = 5 \text{ years}$

$R = 15\%$

$$S.I = \frac{P \times T \times R}{100}$$

$$S.I = \frac{4000 \times 5 \times 15}{100}$$

$$\Rightarrow 200 \times 15$$

$$= 3000 \text{ Rs}$$

3) Given,

$P = 400$

$T = 2\frac{1}{2} = 2.5 \text{ years}$

$R = 18\%$

$$S.I = \frac{400 \times 2.5 \times 12}{100}$$

$$= 4 \times 30$$

$$= 120 \text{ Rs}$$

$$\text{Total amt} \Rightarrow 400 + 120 = 520 \text{ Rs}$$

4) Given,

$$P = 5000$$

$$T = 4 \text{ years}$$

$$SI = 1000$$

$$R = ?$$

$$\frac{5000 \times 4 \times R}{100} = 1000$$

$$200R = 1000$$

$$R = 5\%$$

5) $P = 1 \text{ Rs}$

$$I = 9 \text{ Rs}$$

$$T = 60 \text{ years}$$

$$SI \Rightarrow \frac{1 \times 60 \times R}{100} = 9$$

$$\Rightarrow 60R = 900$$

$$R = \frac{900}{60} = 15\%$$

$$R = 15\% \text{ per annum}$$

6) Given,

$$R = 5\%$$

$$T = 2 \text{ years}$$

$$P = \text{Rs.}$$

$$SI = \frac{P \times 2 \times 5}{100}$$

$$= \frac{10P}{100} = 0.1P = SI$$

$$CI = A - P, \quad A = P \left(1 + \frac{R}{100} \right)^T$$

$$CI = P \left(1 + \frac{R}{100} \right)^T - P$$

$$= P \left[\left(1 + \frac{R}{100} \right)^T - 1 \right]$$

$$= P \left[\left(1 + \frac{5}{100} \right)^2 - 1 \right]$$

$$\begin{array}{r} 105 \\ 105 \\ \hline 525 \\ 105 \times 2 \\ \hline 11025 \end{array}$$

$$CI = P [(1.05)^2 - 1] \Rightarrow$$

$$= P [1.1025 - 1]$$

$$= 0.1025 P$$

Ratio of CI : SI

$$\frac{CI}{SI} = \frac{0.1025P}{0.1P}$$

$$\Rightarrow \frac{1025}{1000} = \frac{41}{40}$$

7) Given,

$$CI = 510$$

$$P = 200$$

$$T = 2 \text{ years}$$

$$R = 4\%$$

By CI & SI relation for 2 years is $\frac{CI}{SI} = \frac{200 + R}{200}$

$$\frac{510}{SI} = \frac{200 + 4\%}{200}$$

$$\frac{510}{SI} = \frac{204}{200}$$

$$\Rightarrow 510 \times 200 = 204 SI$$

$$SI = \frac{510 \times 200}{204}$$
$$= \frac{5 \times 200}{2}$$

$$SI = 500 \text{ Rs}$$

8) Given,

$$P = 400$$

$$A = 441$$

$$T = 2 \text{ years}$$

$$A = P \left(1 + \frac{R}{100} \right)^T$$

$$441 = 400 \left(1 + \frac{R}{100} \right)^2$$

$$\frac{441}{400} = \left(\frac{100 + R}{100} \right)^2$$

$$\sqrt{\frac{441}{400}} = \frac{100 + R}{100}$$

$$\frac{21}{20} = \frac{100 + R}{100}$$

$$2100 = 2000 + 20R$$

$$100 = 20R$$

$$R = 5\%$$

9) Given,

$$P = 800$$

$$R = 5\%$$

$$A = 882$$

$$A = P \left(1 + \frac{R}{100} \right)^T$$

$$882 = 800 \left(1 + \frac{5}{100} \right)^T$$

$$882 = 800 \left(\frac{105}{100} \right)^T$$

$$\frac{882}{800} = \left(\frac{105}{100} \right)^T$$

$$\left(\frac{21}{20} \right)^T = \frac{882}{800}$$

$$T = 2$$

10) Given,

$$T = 3 \text{ years}$$

$$R = 5\%$$

$$CI = 63.05$$

$$CI = A - P$$

$$= P \left(1 + \frac{R}{100} \right)^T - P$$

$$= P \left[\left(1 + \frac{R}{100} \right)^T - 1 \right]$$

$$63.05 = P \left[\left(1 + \frac{5}{100} \right)^3 - 1 \right]$$

$$63.05 = P \left[\left(\frac{105}{100} \right)^3 - 1 \right]$$

$$\begin{array}{r} 105 \\ 105 \\ \hline 525 \\ 000 \times \\ 105 \times \times \\ \hline 11025 \end{array}$$

$$\begin{array}{r} 11025 \\ 105 \\ \hline 55125 \\ 00000 \times \\ 11025 \times \times \\ \hline 1157625 \end{array}$$

$$63.05 = P \left[(1.05)^3 - 1 \right]$$

$$63.05 = P \left[1.157625 - 1 \right]$$

$$63.05 = P \left[0.157625 \right]$$

$$P = \frac{63.05}{0.157625} \times \frac{100000}{100000}$$

$$P = \frac{6305,000}{157,625} = \underline{\underline{400}}$$

Unit - 4

SAT

1) Given : $3A = 4B = 5C$

To find : $A:B:C$

Sol:

$$A:B:C = \frac{1}{3} : \frac{1}{4} : \frac{1}{5} \quad \therefore \text{divid and multiply by 60}$$

$$A:B:C = \frac{60}{3} : \frac{60}{4} : \frac{60}{5}$$

$$A:B:C = 20:15:12$$

$$\frac{aA + bB + cC}{\text{then } A:B:C = \frac{1}{a} : \frac{1}{b} : \frac{1}{c}}$$

2) Given : $A:B = 4:5$
 $B:C = 2:5$

To combin ratios make B value same

multiple $(A:B) \times 2$ & $(B:C) \times 5$

$$\therefore A:B = 4 \times 2 : 5 \times 2 = 8:10$$

$$B:C = 2 \times 5 : 5 \times 5 = 10:25$$

$$\therefore A:B:C = 8:10:25$$

3) A Given: $A:B = 5:6$
 $B:C = 4:7$
 $C:D = 3:1$

Combine sequentially by equating common variable.

$(A:B) \times 2, (B:C) \times 3$

$A:B = 10:12$

$B:C = 12:21$

$A:B:C = 10:12:21$

Now, combine C:D.

$(C:D) \times 7 = 3 \times 7 : 1 \times 7 = 21:7$

$\therefore A:B:C:D = 10:12:21:7$

4) Given: Original ratio = $4:9$

Increase 12 in both,

New ratio : $11:21$

Sol: $\frac{4x+12}{9x+12} = \frac{11}{21}$

$21(4x+12) = 11(9x+12)$

$84x + 252 = 99x + 132$

$15x = 120 \Rightarrow x = 8$

Original no.

$4(8) = 32$

$9(8) = 72$

Sum: $32 + 72$

$= 104$

5) Given : Series : 5, 8, 15

To find : fourth proportional

Sol: $\frac{8}{8} = \frac{15}{x}$

$$\frac{a}{b} = \frac{c}{d}$$

$$x = 24$$

6) & 10) Given : A's investment = 22,500

B's " = 35,000

Profit = 27,600

Sol: $A:B = 22500 : 35000$
 $= 9:14$

$$\text{Profit ratio} = \text{Investment ratio}$$

Total parts = $9 + 14 = 23$

B's share = $\frac{14}{23} \times 27600$

$= 14 \times 1200$

$= 16800 \text{ Rs.}$

$$\frac{11}{10} = \frac{51 + xP}{51 + xP}$$

$$(51 + xP)11 = (51 + xP)10$$

$$51 + xP = 52.5 + xP?$$

$$8 = x \leq 0.01 = x21$$

7) Given: ^{ages} $M \times N = 486$

Ratio = 3:2

Sol: Product = $3x \times 2x$

$$6x^2 = 486$$

$$x^2 = 81$$

$$x = 9$$

$$M = 3(9) = 27, N = 2(9) = 18$$

Sum after 4 years
~~Sum 27 + 18~~
~~= 53 years~~

$$M = 31, N = 22$$

$$31 + 22 = 53 \text{ years}$$

8) Given 60% of A = $\frac{3}{4}$ of B

Given convert 60% = $\frac{60}{100} = \frac{3}{5}$

$$\frac{3}{5} \times A = \frac{3}{4} \times B$$

$$\frac{1}{5} A = \frac{1}{4} B$$

$$\frac{A}{B} = \frac{5}{4}$$

$$A : B = 5 : 4$$

9) Given: 50 paise : Rs 4

$$\boxed{1 \text{ Rs} = 100 \text{ Paise}}$$

Sol: Convert Rs 4 to paise $4 \times 100 = 400 \text{ paise}$

$$\text{Ratio} = 50 : 400 = 1 : 8$$

$$\text{Ratio} = 1 : 8$$

Unit-5

SAQ

1) Given : Speed = 45 km/h

Sol $\boxed{\text{Speed in m/s} = \text{speed in km/h} \times \frac{5}{18}}$

$$\therefore 45 \times \frac{5}{18} = \frac{5 \times 5}{2} = \frac{25}{2} = 12.5 \text{ m/s}$$

2) Given : Distance = 500 meter

Time = 50 sec

Sol : $S = \frac{500}{50} = 10 \text{ m/s}$

$$\boxed{\text{Speed} = \frac{\text{Distance}}{\text{Time}}}$$

$$\text{km/h} = 10 \times \frac{18}{5} = 36 \text{ km/h}$$

3) Given : Distance = 600 m

Time = 5 min

Sol : $5 \times 60 = 300 \text{ sec}$

$$\text{Speed} = \frac{600}{300} = 2 \text{ m/s}$$

$$\boxed{\text{Speed} = \frac{\text{Distance}}{\text{Time}}}$$

$$\text{km/h} = 2 \times \frac{18}{5} = \frac{36}{5} = 7.2 \text{ km/h}$$

4) Given: Case 1, Time = 50 min, Speed = 48 km/L
Case 2, Time = 40 min

Sol: $S_1 \times T_1 = S_2 \times T_2$

$$48 \times 50 = S_2 \times 40$$

$$S_2 = \frac{48 \times 50}{40} = 60 \text{ km/L}$$

5) Given: Train length = 130 m
Speed = 40 km/h

Sol: $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$

$$40 \text{ km/h} = 40 \times \frac{5}{18} = \frac{100}{9} \text{ m/s}$$

$$\text{Time} = \frac{130}{100/9} = \frac{130 \times 9}{100} = 11.7 \text{ sec}$$

6) Given: Train length = 60 m, Platform = 90 m
Speed = 40 km/h

Sol: $\text{Time} = \frac{\text{Train length} + \text{Platform length}}{\text{Speed}}$

$$\text{Total Distance} = 60 + 90 = 150 \text{ m}$$

$$40 \text{ km/h} = 40 \times \frac{5}{18} = \frac{100}{9} \text{ m/s}$$

$$\text{Time} = \frac{150}{100/9} = 1.5 \times 9 = 13.5 \text{ sec}$$

7) Given: Time to pass pole = 15 sec
" " " 100m platform = 25 sec

Sol:
$$\text{Speed} = \frac{L}{15} = \frac{L+100}{25}$$

$$25L = 15(L+100)$$

$$25L = 15L + 1500$$

$$L = 150\text{m}$$

8) Given: Distance = 150m
Time = 25 sec

Sol:
$$\text{Speed} = \frac{150}{25} = 6 \text{ m/s}$$

$$6 \text{ m/s} = 6 \times \frac{18}{5} = 21.6 \text{ km/h}$$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

9) Given: Downstream Speed = 32 km/h = D
Upstream Speed = 28 km/h = U

Sol:
$$\text{Speed in still water} = \frac{D+U}{2}$$

$$\text{Speed} = \frac{32+28}{2} = \frac{60}{2} = 30 \text{ km/h}$$

10) ~~For~~ Given: Train length $L = 100\text{m}$

Speed $= 30\text{km/h}$

Sol: $30\text{km/h} = 30 \times \frac{5}{18}$
 $= \frac{25}{3}\text{m/s}$

Time $= \frac{100}{25/3} = \frac{100 \times 3}{25} = 12\text{sec}$

$$\boxed{\text{Time} = \frac{\text{Distance}}{\text{Speed}}}$$

$$\boxed{\frac{\text{Distance}}{\text{Time}} = \text{Speed}}$$

Given: Distance $= 120\text{m}$

Time $= 32\text{sec}$

$$\text{Speed} = \frac{120}{32} = 3.75\text{m/s}$$

$$\text{Speed} = 3.75 \times \frac{18}{5} = 13.5\text{km/h}$$

1) Given: Downstream speed $= 35\text{km/h}$

Upstream speed $= 25\text{km/h}$

$$\boxed{\text{Speed in still water} = \frac{D+U}{2}}$$

$$\text{Speed} = \frac{35+25}{2}$$

$$= \frac{60}{2} = 30\text{km/h}$$